CVN-69 Milestone 4 Printing Guide

BAMBU P2S PRIMARY · X1C / P1S / A1 · A1 MINI WHERE PRACTICAL · 0.4 MM NOZZLE · PLA

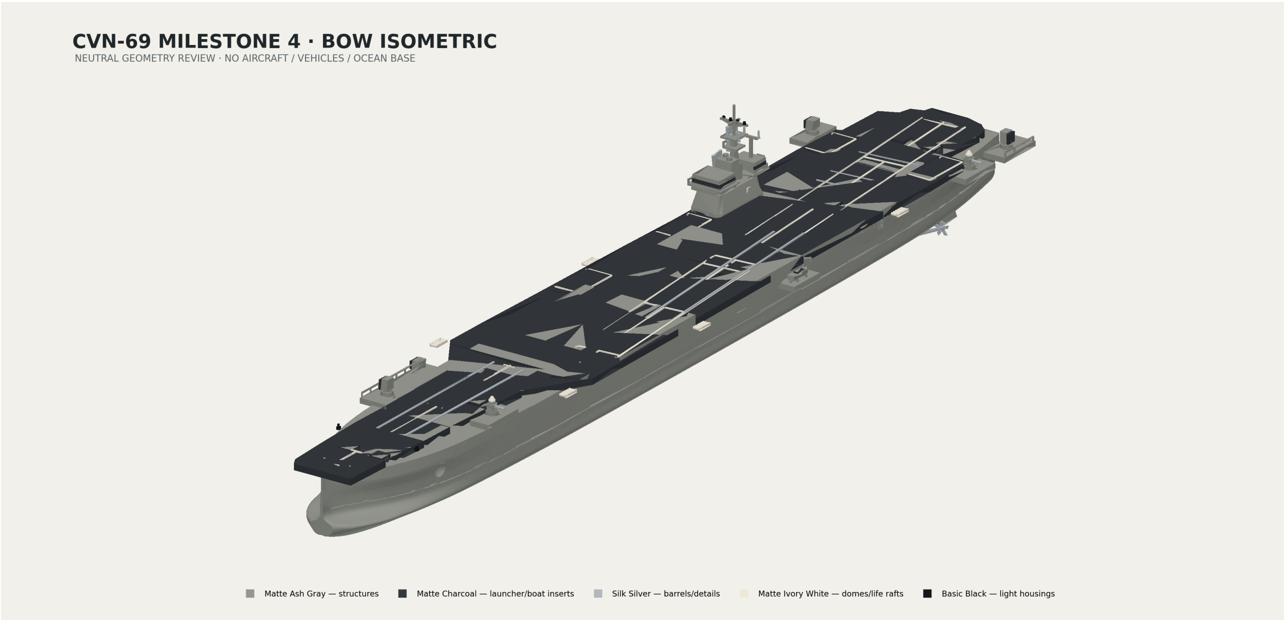
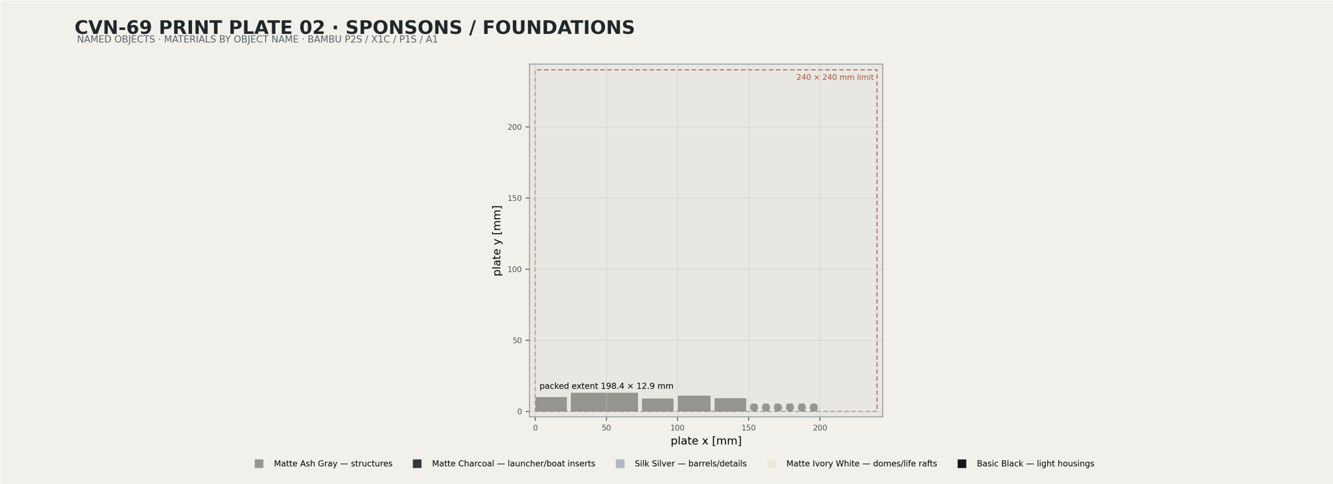


Plate files

3MF	Named-object contents / layer height
Print_Plate_01_Major_Weapons.3mf	launcher/CIWS bodies, faces, domes, barrels · 0.12 mm
Print_Plate_02_Sponsons_Foundations.3mf	five weapon sponsons and six foundations · 0.16/0.12 mm
Print_Plate_03_LifeRafts_Boats.3mf	six life-raft groups, access platform, boat, insert, cradle, davit · 0.16 mm
Print_Plate_04_DeckEdge_Details.3mf	railings, ladders, lights, lockers · 0.12 mm
Weapon_Mount_Interface_Test_Coupon.3mf	actual male/female interface · print first

FDM requirements and orientation

- Do not auto-scale. Exports are millimetres at 1:700 and every production STL rests at z = 0 in its documented orientation.
- Minimum structural wall 1.20 mm; fragile post 0.80 mm; preferred barrel 1.00 mm; raised feature 0.50 mm wide × 0.35 mm high; railing/ladder 0.60 mm.
- Print sponsons and foundations with broad/keyed faces down. Keep launcher faces and fine inserts flat. Avoid supports inside open sockets.
- No part exceeds 240 × 240 × 240 mm. Review/assembly 3MF files are not print plates.

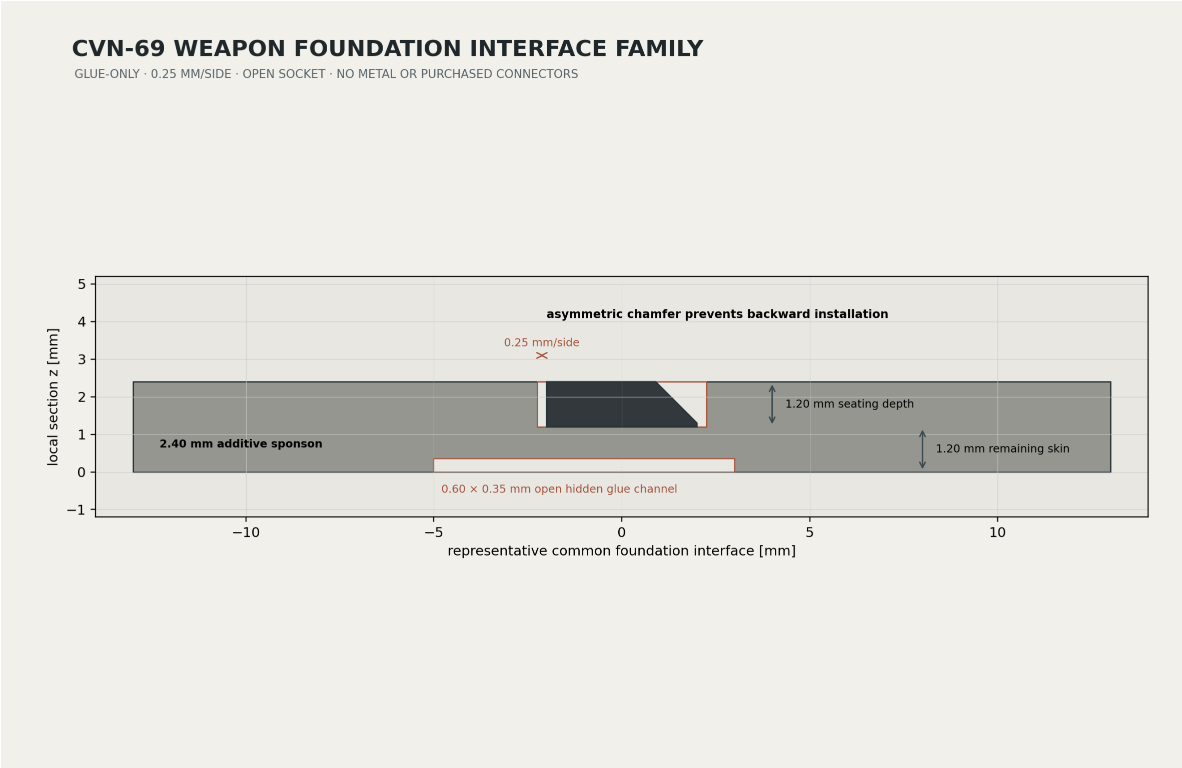


No-paint object materials

Assign filament by object name; no AMS slot is fixed. Separate inserts are preferred over layer-by-layer multicolor waste.

Object	Material
Boat_01_Charcoal_Insert	Bambu PLA Matte Charcoal
Boat_01_Port_Utility	Bambu PLA Matte Ash Gray
Boat_Access_Platform_Port	Bambu PLA Matte Ash Gray
Boat_Cradle_01_Port	Bambu PLA Matte Ash Gray
Boat_Davit_01_Port	Bambu PLA Matte Ash Gray
CIWS_01_Forward_Port_Barrel	Bambu PLA Silk Silver
CIWS_01_Forward_Port_Foundation	Bambu PLA Matte Ash Gray
CIWS_01_Forward_Port_Radar_Dome	Bambu PLA Matte Ivory White
CIWS_01_Forward_Port_Upper_Body	Bambu PLA Matte Ash Gray
CIWS_02_Aft_Port_Barrel	Bambu PLA Silk Silver
CIWS_02_Aft_Port_Foundation	Bambu PLA Matte Ash Gray
CIWS_02_Aft_Port_Radar_Dome	Bambu PLA Matte Ivory White
CIWS_02_Aft_Port_Upper_Body	Bambu PLA Matte Ash Gray
DeckEdge_Access_Ladder_Port	Bambu PLA Matte Ash Gray
DeckEdge_Access_Ladder_Starboard	Bambu PLA Matte Ash Gray
Emergency_Locker_Group_Port	Bambu PLA Matte Ash Gray
Emergency_Locker_Group_Starboard	Bambu PLA Matte Ash Gray
LifeRaft_Group_01_Forward_Port	Bambu PLA Matte Ivory White
LifeRaft_Group_02_Mid_Port	Bambu PLA Matte Ivory White
LifeRaft_Group_03_Aft_Port	Bambu PLA Matte Ivory White
LifeRaft_Group_04_Forward_Starboard	Bambu PLA Matte Ivory White
LifeRaft_Group_05_Mid_Starboard	Bambu PLA Matte Ivory White
LifeRaft_Group_06_Aft_Starboard	Bambu PLA Matte Ivory White
Navigation_Light_Housings	Bambu PLA Basic Black
RAM_01_Forward_Starboard_Foundation	Bambu PLA Matte Ash Gray
RAM_01_Forward_Starboard_Launcher	Bambu PLA Matte Ash Gray
RAM_01_Forward_Starboard_Launcher_Face	Bambu PLA Matte Charcoal
RAM_02_Aft_Port_Foundation	Bambu PLA Matte Ash Gray
RAM_02_Aft_Port_Launcher	Bambu PLA Matte Ash Gray
RAM_02_Aft_Port_Launcher_Face	Bambu PLA Matte Charcoal
Railing_Aft_Port	Bambu PLA Matte Ash Gray
Railing_Forward_Starboard	Bambu PLA Matte Ash Gray
SeaSparrow_01_Forward_Starboard_Foundation	Bambu PLA Matte Ash Gray
SeaSparrow_01_Forward_Starboard_Launcher	Bambu PLA Matte Ash Gray
SeaSparrow_01_Forward_Starboard_Launcher_Face	Bambu PLA Matte Charcoal
SeaSparrow_02_Aft_Starboard_Foundation	Bambu PLA Matte Ash Gray
SeaSparrow_02_Aft_Starboard_Launcher	Bambu PLA Matte Ash Gray
SeaSparrow_02_Aft_Starboard_Launcher_Face	Bambu PLA Matte Charcoal
Sponson_Aft_Port_CIWS	Bambu PLA Matte Ash Gray
Sponson_Aft_Port_RAM	Bambu PLA Matte Ash Gray
Sponson_Aft_Starboard	Bambu PLA Matte Ash Gray
Sponson_Forward_Port	Bambu PLA Matte Ash Gray
Sponson_Forward_Starboard	Bambu PLA Matte Ash Gray

Coupon and first-article acceptance



• Print the coupon using the exact planned printer, material, layer height, and elephant-foot/XY compensation.

• Pass: hand insertion, complete seating, no rocking, no cracked 1.20 mm skin, and the asymmetric key cannot reverse.

• Fail: force fit, incomplete seating, more than 0.10 mm vertical error, visible lean over 0.30°, or glue trapped below the key.

• Correct calibration and reprint. Never scale the production ship or sand the key into symmetry.